Microeconometrics Using Stata

Simulation

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Exercise 1

Exercise 1

Exercise 1

Give the commands **set obs 100** and **generate u = runiform()** and **summarize**. Then, give command **clear** and repeat the initial commands. Did you get the same result? If not, explain why, and adapt your code to get the same results.

Exercise 2

Exercise 2

Exercise 2

Using the normal generator, generate a random draw from a $50-50$ scale mixture of $N(1,1)$ and $N(1,3^2)$ distributions. Repeat the exercise with the $N(1,3^2)$ component replaced by $N(3,1)$. For both cases, display the features of the generated data by using a kernel density plot.

Exercise 3

Exercise 3

Exercise 3

Generate $1,000$ observations from the $F(5,10)$ distribution. Use **rchi2()** to obtain draws from the $\chi^2(5)$ and the $\chi^2(10)$ distributions. Compare the sample moments with their theoretical counterparts.

Exercise 4

Exercise 4

Exercise 4

Make $1,000$ draws from the $N(6, 2^2)$ distribution by making a transformation of draws from $N(0, 1)$ and then making the transformation $Y = \mu + \sigma Z$.

Exercise 5

Exercise 5

Exercise 5

Generate $1,000$ draws from the $t(6)$ distribution, which has a mean of 0 and a variance of 4. Compare your results with those from **exercise 4**.

Exercise 6

Exercise 6

Exercise 6

Generate a large sample from the $N(\mu = 1, \sigma^2 = 1)$ distribution and estimate σ/μ , the coefficient of variation (**CV**). Verify that the sample estimate is a consistent estimate.

Exercise 7

Exercise 7

Exercise 7

Generate a draw from a multivariate normal distribution, $N(\mu, \Sigma = \mathbf{LL}')$, with $\mu' = [0x0x0]$ and

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & \sqrt{3} & 0 \\ 0 & \sqrt{3} & \sqrt{6} \end{bmatrix}, \text{or} \quad \Sigma = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 4 & 3 \\ 0 & 3 & 9 \end{bmatrix} \tag{1}$$

using transformations based on this Cholesky decomposition. Compare your results with those based on using the **drawnorm** command.

Exercise 8

Exercise 8

Let s denote the sample estimate of σ and $\bar{x}$ denote the sample estimate of μ. The **CV** σ/μ , which is the ratio of the standard deviation to the mean, is a dimensionless measure of dispersion. The asymptotic distribution of the sample **CV** $s/\bar{x}$ is

$$N\left\{\frac{\sigma}{\mu},(N-2)^{-\frac{1}{2}}\left(\frac{\sigma}{\mu}\right)^2\left[0.5+\left(\frac{\sigma}{\mu}\right)^2\right]\right\} \tag{2}$$

see **Miller (1991)**. For $N=25$, using either **simulate** or **postfile**, compare the Monte Carlo and asymptotic variance of the sample **CV** with the following specification of the **DGP**: $x \sim N(\mu,\sigma^2)$ with three different values of **CV** $=0.1, 0.33$, and 0.67.

Exercise 9

Exercise 9

Exercise 9

It is suspected that making draws from the truncated normal using the method given in **section 5.4.4** may not work well when sampling from the extreme tails of the normal. Using different truncation points, check this suggestion.

Exercise 10

Exercise 10

Repeat the example of **section 5.6.1** (OLS with χ^2 errors), now using the **postfile** command. Use **postfile** to save the estimated slope coefficient, standard error, the t statistic for $H_0 : \beta = 2$, and an indicator for whether H_0 is rejected at the 0.05 level in a Stata file named **simresults**. The template program is as follows (see template):

References I

Cameron, A. C. and Trivedi, P. K. (2022a).
Microeconometrics Using Stata: Volume I: Cross-Sectional and Panel Regression Methods.
Stata Press.

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Microeconometrics Using Stata: Volume II: Nonlinear Models and Causal Inference Methods.
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```
* Postfile and post example: repeat OLS with chi-squared errors example
clear
set seed 10101
program simbypost
    version 17
    tempname simfile
    postfile `simfile´ b2 se2 t2 reject2 p2 using simresults, replace
    quietly {
       forvalues i = 1/1000 {
         drop _all
         set obs 150
         generate x = rchi2(1)
         generate y = 1 + 2*x + rchi2(1) - 1   // demeaned chi^2 error
         regress y x
         scalar b2 =_b[x]
         scalar se2 = _se[x]
         scalar t2 = (_b[x]-2)/_se[x]
         scalar reject2 = abs(t2) > invttail(1000-2,.025)
         scalar p2 = 2*ttail(1000-2,abs(t2))
         post `simfile´ (b2) (se2) (t2) (reject2) (p2)
      }
    }
    postclose `simfile´
end
simbypost
use simresults, clear
summarize
```